

# From Data to Diagram: Learning Chart Types Through Interactive Instructional Comics

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Learning and exploring data visualization requires **interaction**, such as manipulating charts, switching views, or accessing details on demand.

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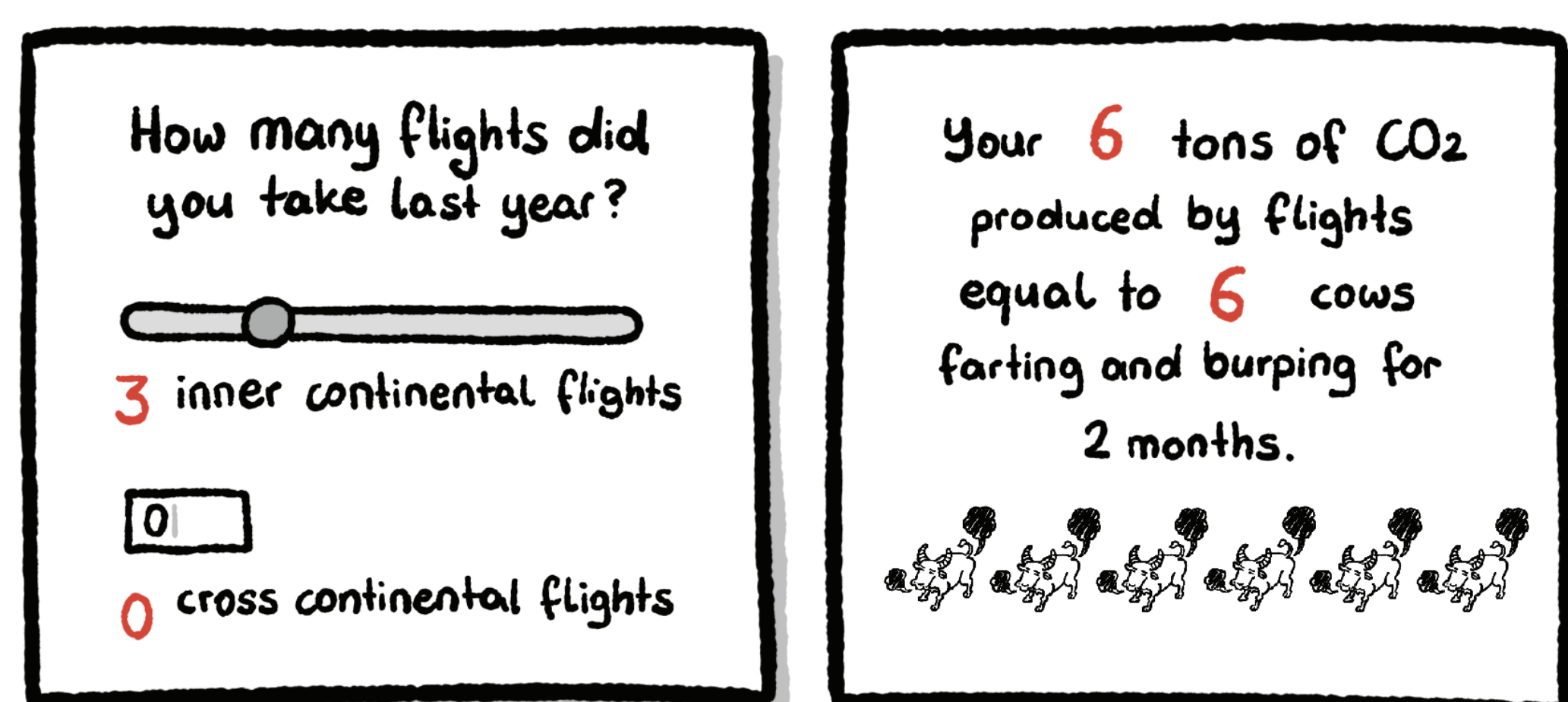
Educators strongly advocate for using meaningful **personalization** and own data when learning visualization.



Premise: Expanding educational comics through interactivity and personalization

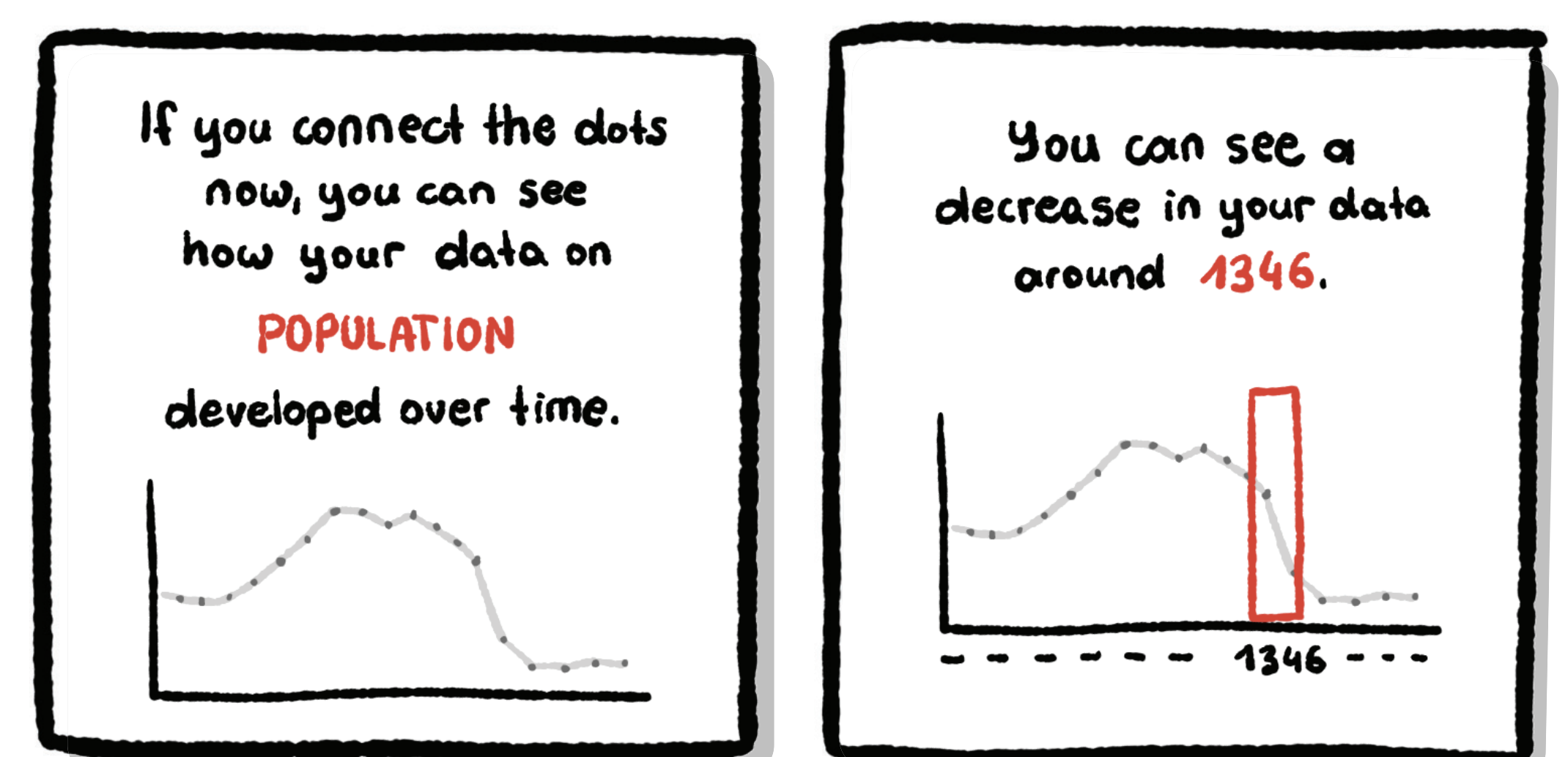
Currently, the extent to which interactive (data) comics can adapt to user content is largely limited to sliders and input fields [1].

If personalization were added to an explanatory comic on e.g., how to read a certain chart type, it could only provide superficial commentary.



(adapted from the interactive data comic "Personal CO2 Emissions" by Benjamin Bach) [3]

BUT



This works reasonably well for data comics, which focus on specific datasets or have a tightly scoped context.

However, with the advent of LLMs, such a comic could incorporate reasoning and context, allowing for commentary like:

"The year 1346 shows a sharp drop in your population data, likely due to the outbreak of the Black Plague."

## DESIGNING AN LLM-SUPPORTED, ADAPTIVE, EXPLANATORY COMIC

The comic follows a predefined narrative structure, e.g.:

Anatomy  
↓  
Construction  
↓  
Visual Patterns  
↓  
Pitfalls

(based on [2])

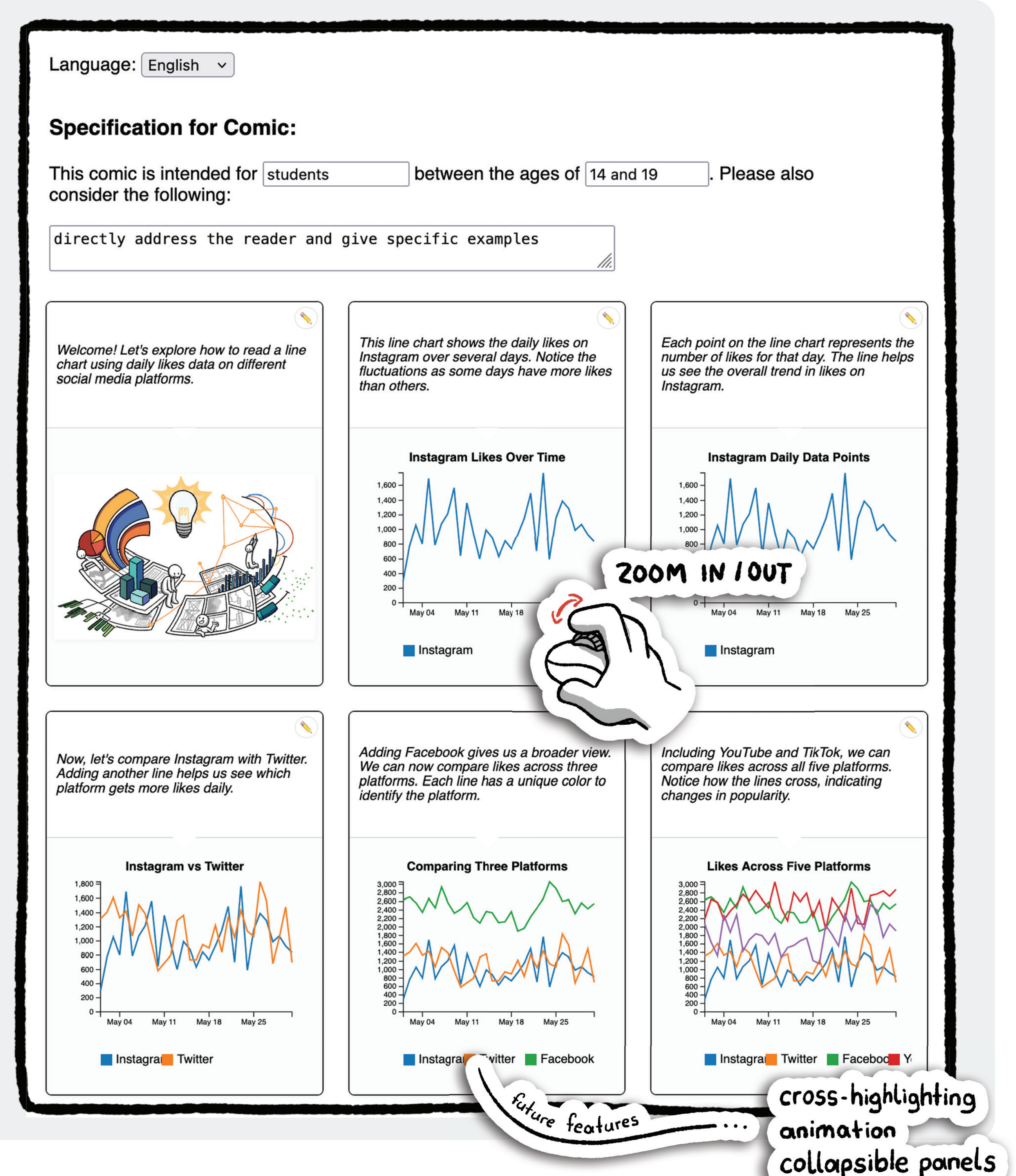
This ensures educational quality and consistency across user data uploads.

The user uploads a CSV and provides some context...



"These are my daily likes on social media platforms"

...and the LLM reads the dataset and context and generates a customized explanatory comic along the structure.



[1] Wang, Z., Romat, H., Chevalier, F., Riche, N. H., Murray-Rust, D., & Bach, B. (2022). Interactive Data Comics. IEEE Transactions on Visualization and Computer Graphics, 28(1), 944–954. <https://doi.org/10.1109/TVCG.2021.3114849>

[2] Wang, Z., Sundin, L., Murray-Rust, D., & Bach, B. (2020). Cheat Sheets for Data Visualization Techniques. Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, 1–13. <https://doi.org/10.1145/3313831.3376271>

[3] Interactive Data Comic "Personal CO2 Emissions" by Benjamin Bach: <https://hugoromat.github.io/interactiveComics/library/dist/CO2.html>